Vertically Integrated Industries

Initial takeaways on minerals, metals, manufacturing and chemistry

Smith Point Capital theme memo, v3, October 28, 2026. Drawn from our primer (v10.1), its sources and new research on Chemify and Warp Sciences. Company statements are labeled as claims.

What this theme is

This theme covers companies that make physical things in their own plants and use their own software and AI to make them faster, and with fewer skilled people, than older competitors. Their revenue comes from the part, metal or molecule they ship. We call them vertically integrated because one company owns the whole job, from taking the order to shipping the product.

Hadrian is the clearest example. It machines metal parts for rockets, missiles, satellites, jets and submarines in automated factories in California and Arizona. Its in-house software handles quoting, programming the machines, scheduling and quality checks. A customer sends a drawing and gets parts back. Lockheed liked the result enough to put Hadrian's cells inside its own missile plant.

Picture the supply chain as a ladder. Companies mine ore, then refine it into pure metals and products like magnets. Distributors cut metal to size. Fabrication shops cut, bend and weld it into frames and enclosures. Precision shops make tight-tolerance parts by computer-controlled machining (CNC), metal 3D printing or casting, mostly for aerospace and defense. Chemistry sits alongside: making the molecules that drug and materials programs need. At the top sit the primes, the few big contractors (Lockheed, RTX, Northrop, General Dynamics, Boeing) that sell finished weapons and aircraft to the government.

Why now? China has shown it will cut off critical minerals and magnets. The Pentagon needs far more missiles and ships than its suppliers can make. Washington is lending, buying and taking equity stakes to get US production built. And the owners of America's small machine shops are retiring with no one to replace them.

1. Key takeaways

Five steps, each building on the last: demand, capex risk, margin, AI's edge, incumbents.

1. Demand for US production is large and mostly lasting, but much of the money is conditional.

Tailwind	Key numbers	Lasting or cyclical	Who it helps most
China's hold on refining and magnets	Leads refining in 19 of 20 strategic minerals at about 70% share (IEA); about 94% of rare-earth magnets (Mining.com); US fully import-reliant for 16 minerals (USGS)	Lasting. China's pause on export controls ends Nov 10 and Nov 27, 2026	Magnet and metal makers; parts makers that need secure inputs
Munitions and shipbuilding	$49.5B for 10 munitions in the FY27 request, 8.5x the past decade's average, but 82% depends on an unpassed reconciliation bill (CNAS); $900M of Navy money for submarine-part plants (Breaking Defense)	Need is lasting; funding timing is cyclical	Precision parts and structures

Tailwind	Key numbers	Lasting or cyclical	Who it helps most
Multi-year buying in law	FY26 defense bill allows multi-year munitions buys and purchase commitments of up to 10 years with advance payments (FY26 NDAA)	Lasting law; each year still needs appropriations	Producers with a plant already running
Government lending and equity	Pentagon OSC (Office of Strategic Capital) loans of $620M to Vulcan and $500M (conditional) to Phoenix, about $20B requested for FY27 (Mintz); DOE loan office has over $289B of authority (DOE); $65B of public minerals money worldwide in 2025 (IEA)	Mixed. Large, but conditional and political	Capital-heavy producers that can carry debt
Labor gap	3.8M manufacturing jobs to fill by 2033, 1.9M may go unfilled (Deloitte); 83% of 17,156 US machine shops have under 20 staff (Census)	Lasting	Automated producers and buyers of old shops
Drug-chemistry onshoring	BIOSECURE became law in Dec 2025; 79% of 124 biotechs surveyed had a China contract manufacturer (BIO); replacing them could take up to eight years (industry estimate)	Lasting policy, slow enforcement	US chemistry labs
Tariffs	50%, 25% and 10% tiers on metal goods since April 2026 (White & Case); Nucor's margin rose from 14% to 20%	Cyclical. Set by proclamation	Domestic mills first
Reshoring and data centers	Fabricated-metal shipments rose from $471.5B in 2023 to $490.4B in 2025 (Census); only 14 of 64 tariff-linked reshoring announcements had turned into projects by mid-2026 (Reshoring Initiative)	Partly cyclical	Fabricators

- Security and labor will keep buyers paying for US capacity for a decade. Budgets and tariffs explain why 2026-27 feels hot, and they could cool.
- Against us: the Fed raised rates to 3.75-4.00% in September, and manufacturing construction was down 21.9% year on year by May 2026 (Federal Reserve, ENR). Bonus depreciation helps profitable incumbents buy machines.

2. To serve that demand, a factory has to spend its capital before it wins the contracts. This is the central risk in the theme.

- A buyer won't qualify you or sign a volume contract until the plant, machines and certifications exist. Hadrian's CEO says "no one's going to give you a production contract unless you've already started executing," and capex runs about two years ahead of revenue (claim). Qualification takes 9-12 months at Raytheon and one to two years at Northrop (expert calls).
- Examples: Hadrian committed about $200M to its Mesa, Arizona plant (Plant Services). Divergent is filling a 430,000-square-foot Long Beach factory sized mostly for missile work still waiting on the FY27 budget. Vulcan is building a 10,000-ton magnet plant with no commercial buyer named. Phoenix announced a $375M Rochester plant with one $100M automaker contract. Chemify's second lab is part of an £89.9M expansion (The Scotsman).
- Equity spent on a building earns building-like returns. A hardware company at Series B often has about $10M of revenue and break-even gross margins (Lowercarbon).
- The best companies move the risk to someone else: government loans (Vulcan's $620M), offtakes signed before the build (the Pentagon agreed to buy all of MP Materials' next magnet

plant output for 10 years, with a price floor), customer-hosted cells (Lockheed and Hadrian, 7 years), bank debt once revenue exists (Hadrian's $360M credit line), leases (Senra's second plant), franchising (Isembard) and buying plants that are already certified (Amca).

3. The companies that do build, own the output and get approved keep the margin. Buyers reward them for speed more than price.

- Owned plants earn more than partner networks. Proto Labs makes 49% gross margin in its own plants and 31% on work it farms out (Protolabs). Approved incumbents earn well: Howmet's engine unit runs a 37.7% EBITDA margin and Carpenter 32.1%.
- Buyers pay for speed on prototypes, spares and urgent orders. "Cost is secondary. Cost is driven usually by schedule" (Northrop director). A Northrop engineer would pay 20-40% more to avoid a delay. The premium fades at volume, and buyers cap each supplier's share. Hadrian holds 2-3% of one Boeing unit's precision spend (expert calls).
- Labor is under 10% of cost of goods in a new automated plant (claim, Drew Baglino), so the pitch is "we deliver when your current shop can't."

4. AI's real edge is in the office work around the machine and in processes that run nonstop. Nobody has shown it making each unit cheaper as volume grows.

- AI speeds quoting, programming and instructions. Arda cut assembly-instruction writing at Anduril from about 10 hours to 15 minutes (claim), and Hadrian's order-to-first-part time fell about 50% over two years (former employee).
- The data advantage is strongest where every run is measured: Freeform's printers checking each layer, robotic chemistry labs logging failed reactions. A Boeing strategy head puts net savings from automation, after paying for the machines, at 20-30% at best.
- That points to software paid on output. In a sold-out process plant each extra point of yield is near-pure profit, and European plants average about 74% utilization against the 80-85% they need (FTI Consulting).

5. The incumbents are automating and buying, so the winners need several buyers and approvals that are hard to copy.

- The primes decide who is qualified and cap share. Precision Castparts' share of Boeing machining fell from about 50% to 30%, and it is automating (expert call). Pharma is building its own AI labs: Eli Lilly and NVIDIA plan up to $1B for one (Lightspeed).
- The majors buy rather than build: BHP Ventures backs Mariana, Rio paid $6.7B for Arcadium Lithium, and RTX Ventures led Radical AI's round.
- Daedalus, an AI-run CNC factory in Germany, went insolvent in August 2026, and Markforged fell from $2.1B to a $42.5M sale (3D Printing Journal).

2. Investable now

These companies have revenue or paying partners, a stage and price we can work with (under about $10B), and a plausible answer to the capex problem. Ranked: (1) Hadrian, (2) Isembard, (3) Lumaril, (4)

Amca, (5) Senra Systems, (6) Divergent, (7) Freeform, (8) Chemify, (9) SendCutSend, (10) Machina Labs, (11) Vulcan Elements.

Hadrian ranks first on proof: primes qualify it and pay for its parts. Isembard and Lumaril follow because they need the least capital and are raising now. Vulcan is last because it carries the most plant risk.

A. Precision and defense parts

Owned vs hybrid

These companies split into two models. Owned companies run their own factories and machines, keep the whole margin and all the data, and carry the full cost of building before the contracts come. Hybrid companies mix their own plants with capacity someone else paid for. They keep less of each sale but need far less capital.

Company	Model	Who pays for the machines	Who owns the customer	Who carries the capex-before-contracts risk	What it means for margins and returns
Hadrian	Owned, with hosted cells (our primer labels it hybrid)	Hadrian for its own plants; the Navy co-funds the submarine plants; cells at Lockheed sit in Lockheed's building	Hadrian	Mostly Hadrian, shared with the Navy and Lockheed	Full margin and data; returns depend on filling very expensive plants
Divergent	Owned	Divergent (its own printers and Long Beach factory)	Divergent	Divergent alone	Full margin and data; most exposed to FY27 munitions timing
Freeform	Owned	Freeform (its own printers)	Freeform	Freeform, at smaller scale	Same model as Divergent at a much lower entry price
Amca	Owned, by acquisition	Already paid for by the shops it buys	Amca, which inherits the shops' prime relationships	Little build risk; the risk moves to purchase price and debt	Returns come from buying well and lifting margins at existing plants
Isembard	Hybrid (4 owned, 10 franchised sites, claim)	Franchisees for most sites	Isembard	Mostly franchisees	Smaller share of each part's margin, far less capital; works only if quality holds across sites. The split with franchisees is not public

How it ties to takeaways 2 and 3: every company here sells speed into programs that can last 50 years, and the model decides who pays for the plant. Incumbents leave room because the lower tiers are huge and untracked. The Pentagon relies on more than 200,000 suppliers and has country-of-origin data for fewer than 10% of F-35 sub-tier suppliers (GAO).

Hadrian ($1.37B Series D, August 2026, at $7.87B; anchored by JPMorgan's Strategic Investment Group; about 643 staff)

- Overview and use cases: Hadrian makes CNC-machined metal parts for rockets, missiles, satellites, jets and, from 2026, Virginia- and Columbia-class submarines. Its factories in Torrance, California and Mesa, Arizona run on its own Opus software, and it also runs cells inside customer plants such as Lockheed's. Customers include Lockheed, RTX, Anduril and the Navy.
- Thesis: Hadrian has more proof than anyone in the theme that primes will qualify and pay a new producer, and it has shown the capex playbook in takeaway 2: Navy co-investment, a Lockheed-hosted cell deal and a $360M credit line (ABL Advisor). Our help would be introductions to commercial buyers outside defense.
- Areas to diligence: A former employee called it "not … a fantastically profitable business," so we need margins at $7.87B. We would test whether Opus lowers unit cost over time and how Hadrian grows past the 25-30% share buyers allow. Its CEO says some Chinese parts land at about a third of his cost (claim). Access to an allocation is to confirm.

Isembard ($50M Series A, March 2026, led by USV; claims 14 factories)

- Overview and use cases: Isembard machines precision parts from aluminum, steel, titanium and Inconel (a heat-resistant nickel alloy) for jet engines, drones, satellite antennas and robots. Named customers include Anduril, Babcock, Tekever and Oxford Space Systems. It runs owned and franchised factories on one system, MasonOS, from a London headquarters with sites in the US, Germany, France and Ukraine. It claims 10x revenue growth year on year.
- Thesis: Franchisees pay for the machines while Isembard supplies software, orders and standards. That is the lightest-capex answer to takeaway 2. It claims it cut one customer's lead time from about 10 weeks to under 14 days at the same price. In September 2026 it said it expects to raise significant additional funding.
- Areas to diligence: We would ask what a franchisee earns, what Isembard keeps and how fast a site turns a profit. A Thales Alenia Space procurement head doubted primes will accept quality across franchised sites.

Amca ($300M Series B at over $1B, led by Caffeinated Capital and Lightspeed; revenue to confirm)

- Overview and use cases: Amca buys certified aerospace and defense component makers and modernizes them. It owns eight plants, including Electrocube in Pomona, California, that make hydraulic and electrical parts, switches, valves, transformers and power supplies for the F-35, F-16, F/A-18, Mk-48 torpedo and M1 Abrams. Its AI platform, RAPID, handles engineering and requalification. It claims an Army Materiel Command contract and work with over a dozen primes.
- Thesis: Amca buys capex and qualification that already exist, often for single-source parts primes can't easily move, then adds software to lift margins. A buy-and-improve platform suits our operating and partnership support.

- Areas to diligence: Revenue, EBITDA and the price paid per plant are to confirm. One roll-up founder said they buy shops at about seven times EBITDA (Re:Build). We would test whether RAPID lifts margins, how much debt sits on the plants, and whether key engineers stay.

Divergent ($290M Series E, September 2025, at $2.3B; over $1B raised; about 500 staff)

- Overview and use cases: Divergent designs and 3D-prints metal structures for missiles, aircraft and cars, then assembles them with robots and no fixed tooling. It builds its own printers and charges per part. Named customers include Lockheed, RTX, General Atomics, Saab, Bugatti and McLaren. It is a second source for Raytheon's Tomahawk midbody, with production aimed for early 2027.
- Thesis: It owns everything from design software to factory, sells into both defense and cars, and claims 2025 revenue grew more than 5x. The two-market mix hedges budget risk.
- Areas to diligence: Long Beach is the biggest capex-before-contract bet in our set, since about 80% is sized for defense volume (CEO). Cost parity with casting is only the CEO's claim. A newer priced round is to confirm.

Freeform ($67M Series B, February 2026; about $179M post-money per PitchBook)

- Overview and use cases: Freeform 3D-prints metal parts that carry fluid and heat, like heat exchangers, manifolds and rocket-engine injectors, for space, defense, energy and chip-equipment customers (claim). Its own multi-laser printers use cameras and AI to correct each layer as it prints. It sells the parts and keeps the printers. It reports hundreds of parts a week and has a Defense Innovation Unit contract.
- Thesis: Freeform has the best data story in manufacturing, because every layer is measured (takeaway 4), at the cheapest entry in this area.
- Areas to diligence: Customers are undisclosed, so we need reference calls. We haven't seen yield trends over time, and whether its large new printer is running is unconfirmed.

B. Fabrication and assembly

How it ties to takeaways 2 and 3: these companies sell speed on high-mix work (many small, different orders) from plants that cost tens of millions. Senra and SendCutSend own their plants; Machina is hybrid, because it also places robot cells at customer sites. Incumbent shops are small, aging and short of programmers, and their machines cut metal only about 34% of the time against a 70% target (industry survey of 100+ plants).

Senra Systems ($65M Series B, July 2026, co-led by Lowercarbon and Interlagos; about 190 staff)

- Overview and use cases: Senra makes wire harnesses and cable assemblies, the bundles of wires and connectors that carry power and signals through aircraft, satellites and defense systems. It builds them in two plants in Redondo Beach and Cypress, California. Its software, Amp, turns a customer's design into a digital model and step-by-step instructions for technicians. It claims it cut harness lead time from six months to two weeks, and says 99% of harnesses pass inspection the first time.

- Thesis: Harnesses are still made by hand, and a late one holds up the whole aircraft or missile. Senra leased its second plant and opened it within months, the light-capex approach we want. It just raised, so this is a relationship to build for the next round.
- Areas to diligence: We would check the yield and lead-time claims with customers, gross margin by program and how concentrated repeat revenue is.

SendCutSend (Series B at $2.1B, October 2026, after a $1.01B round in May 2026)

- Overview and use cases: SendCutSend cuts, bends and finishes custom sheet-metal parts that customers order online, in plants it owns. It handles about 32,000 orders a month and has been profitable for eight years.
- Thesis: It proves an owned fabrication business can be profitable at scale, and it was bootstrapped, so earnings paid for the plants.
- Areas to diligence: The price doubled in five months, and the $2.1B figure comes from a single trade report, so it is to confirm. Customers tend to "outgrow" online fabricators, and the founder says he beats China on only about 20-25% of his work (claim).

Machina Labs ($124M Series C, February 2026)

- Overview and use cases: Machina shapes sheet metal into missile airframes and aircraft and car panels with pairs of robot arms, skipping the costly dies a stamping press needs. It claims its first plant will make thousands of missile airframes a year.
- Thesis: A new die takes months and a lot of money, so skipping it suits defense programs whose designs change often. The founder says Machina won't start a project unless the customer commits dollars, which is good capex discipline.
- Areas to diligence: Each part takes 30 minutes to hours, against seconds on a press (MMS). We would test where it wins at volume, verify the claimed 50x lead-time gain, and split paid production from development work.

C. Software that runs plants

How it ties to takeaways 2 and 4: this is the way to back the theme without carrying a plant. Rockwell, Siemens and the process-control vendors (Honeywell, AspenTech, Emerson) sell hardware first or advise operators, which leaves room for software that runs the plant and is paid on what it adds.

Lumaril ($12M seed led by Lux; a WSJ Pro headline reports a $66M round at $400M)

- Overview and use cases: Lumaril connects to a process plant's existing controls, adds cheap audio and video sensors, builds a live model of the plant and adjusts setpoints itself, through a safety layer, to raise throughput and yield. A setpoint is a target such as a temperature or flow rate. It says "We're paid on the production we add" (claim). The founders built Palantir Foundry, Palantir's commercial data platform.
- Thesis: Lumaril earns like a producer with no capex, and because it runs the plant directly it builds the data a plant owner would. The buyers are large industrial enterprises, where our

partnership help counts. It is early, but we rank it third because the model fits us best and a round appears to be under way.

- Areas to diligence: Customers and results aren't public, so we need one plant's before-and-after yield. We would ask how the outcome-pricing baseline is set and how a plant's safety team approved software that adjusts the plant on its own. The round size comes only from a paywalled headline.

D. AI-run chemistry labs

How it ties to takeaways 1, 2 and 4: Chemify and Warp Sciences are the only chemistry companies we want in the theme. Both build robotic labs that make small batches of new molecules for drug and materials programs and learn from every run. Drug companies send most of this work offshore, largely to China, and the two biggest offshore providers are worth close to $100B combined (Lightspeed). BIOSECURE and the Pentagon's June 2026 listing of the largest provider as a Chinese military company push buyers toward a US option. Chemify is investable now. Warp is in section 3 because it launched last week with no customers yet.

Chemify (Series B of £51.9M, about $70M, co-led by Wing and Insight Partners; about £110M raised including grants; valuation to confirm)

- Overview and use cases: Chemify, a University of Glasgow spin-out founded by chemist Lee Cronin, turns molecule designs into code that robots run to make the compound. It calls this "chemputation," built on its own chemistry programming language and a large library of tested reactions (Insight Partners). Its first automated lab, the Chemifarm, opened in Glasgow in June 2025 and makes molecules at milligram-to-gram scale (claim, SCI). It mainly sells discovery chemistry to pharma. Named partners include Sanacor (heart disease) and the Gates Foundation (tuberculosis), and Cronin claims it works with eight of the top 20 pharma companies (SCI). Staff are about 152.
- Thesis: Chemify owns the lab, the language and the reaction data, so customers' experiments improve its system. It pushed part of its capex onto government, with £22M of Scottish and UK grants toward its second lab. It plans a US lab, where our go-to-market and pharma introductions could help.
- Areas to diligence: Revenue is undisclosed, and the "eight of the top 20" claim may include pilots. Milligram-to-gram output means this is research services for now. We would ask whether the US lab is funded and how it competes with pharma's in-house robotic labs.

E. Rare-earth magnets

How it ties to takeaways 1 and 2: China's hold on magnets is the most lasting limit in the theme, but these plants carry the most capex before contracts. MP, the only scaled US producer, still lost $85.9M on $224.4M of revenue in 2025 (MP Materials). We would back at most one company here, and only where government debt pays for the plant and a buyer already pays.

Vulcan Elements ($65M Series A, led by Altimeter; $620M OSC loan)

- Overview and use cases: Vulcan makes NdFeB magnets (named for neodymium, iron and boron) for drones, chip tools, robots and cars from non-Chinese material. It buys refined oxides from ReElement, has a supply deal with Ucore from 2027 and is building a 10,000-ton plant in Benson, North Carolina. The Army chose it for its SkyFoundry drone program.
- Thesis: A defense buyer already orders its product and government debt pays for the plant, which is the right capex structure. It claims its pilot output is sold out a year ahead.
- Areas to diligence: Federal awards total only $5.7M across seven contracts (USAspending). We would test unit cost against Chinese magnets and whether commercial buyers sign without subsidy.

Recommendation

Hadrian, Isembard and Lumaril are the most actionable. Hadrian has already solved the capex problem with Navy money, a Lockheed-hosted deal and bank debt; the question is price and access at $7.87B. Isembard has the lightest capex model and says it expects to raise soon. Lumaril fits an enterprise software fund best, paid on output with no plant. Amca and Senra are close behind.

3. Not investable now

Company or layer	Why not now	What would change our mind
Warp Sciences ($215M Series A, Lightspeed; $865M post per CEO)	Too early. Launched October 22, 2026 with design partners only and no revenue, at a high launch price	Named paying pharma customers and a lab running at volume
Phoenix Tailings ($116.6M Series B in total)	Too capital-heavy for our check. Plans to grow about 15x in two years; OSC loan conditional; no public mine supply	A final OSC loan, signed long-term supply and a round sized for us
Mariana Minerals ($310M Series B, Khosla)	Too capital-heavy. Owns a Utah copper mine and refinery; no copper buyer named	A named offtake and a round we can join at a meaningful size
1872 ($15M seed)	Too early. No customer evidence; welding AI comes from a partner	A first repeat customer and software it owns
Arda ($160M Series A at $1.5B)	One named customer (Anduril) at a high price	Several factories beyond Anduril paying for it
Radical AI ($55M, RTX Ventures)	No product revenue; a new alloy still needs one to five years of buyer qualification	A signed supply contract with a defense buyer
CuspAI ($450M Series B)	No product revenue; priced on promise	Materials in a customer's product
Periodic Labs ($300M seed), Lila Sciences	Discovery labs valued in the billions before selling anything	A paid commercial product
Earth AI	Exploration. US mines take 15.7 to 19 years from discovery to production; no formal resource estimate yet	A published resource estimate and a buyer for the deposit
Mind Robotics ($3.4B)	One main customer, Rivian, which owns 32.1% of it	Paying customers beyond Rivian
Metal distribution	Incumbents like Reliance are gaining share without AI; value sits in inventory and credit	A distributor showing clearly better margins from software

Company or layer	Why not now	What would change our mind
Greenfield refineries funded with equity	One giant bet better suited to government debt	Debt and offtakes covering most of the build
Bulk and specialty chemical producers	Commodity price fights and long regulatory reviews	Our chemistry exposure comes through Chemify and Warp
Companies valued above about $10B	We can't own enough to matter	A secondary at a price that works

4. What we would underwrite

We back companies that own their output, get paid for speed, and have moved the cost of their plants onto someone else. Five tests:

1. Committed demand before the build: a signed offtake, a prepayment, a customer-hosted deal or a qualified buyer already ordering.
2. Debt, government loans or grants, customer money, leases, franchisees or an acquired plant pay for most of the capex, so our equity funds software, people and ramp-up.
3. It owns production or the outcome of production, through its own floor, a franchise network on its software, or pricing tied to the output it adds.
4. It sells to more than one big buyer and holds approvals that take years to repeat.
5. It can show with numbers that its software improves yield, lead time, scrap or cost over time.